Supporting Information: Molecular engineering of semi-transparent non-fullerene acceptors for quantum-enhanced agrivoltaics

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1 OPV materials synthesis protocols

1.1 PM6 derivative synthesis

The PM6 derivative (poly[(2,6-(4,8-bis(5-(2-ethylhexyl-3-fluoro)thiophen-2-yl)-benzo[1,2-b:4,5-b']dithiophene))-alt-(5,5-(1',3'-di-2-thienyl-5',7'-bis(2-ethylhexyl)benzo[1',2'-c:4',5'-c']dithiophene-4,8-dione))]) was synthesized via Stille coupling polymerization.

Monomer preparation.

1. **BDD-T monomer:** 4,8-Bis(5-(2-ethylhexyl-3-fluoro)thiophen-2-yl)benzo[1,2-b:4,5-b']dithiophene was synthesized via Pd-catalyzed C-H activation. Yield: 78 %. ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.65 (d, J = 4.2 Hz, 2H), 7.32 (d, J = 4.2 Hz, 2H), 2.85 (d, J = 6.8 Hz, 4H), 1.72–1.25 (m, 36H), 0.92 (t, J = 7.0 Hz, 12H).

2. **DTBDT monomer:** 5,5'-(1',3'-Di-2-thienyl-5',7'-bis(2-ethylhexyl)benzo[1',2'-c:4',5'-c']dithiophene-4,8-dione) was prepared via sequential lithiation and quenching. Yield: 65 %. ^{1}H NMR (500 MHz, $CDCl_3$): δ 8.92 (d, J = 3.8 Hz, 2H), 7.58 (d, J = 3.8 Hz, 2H), 2.95 (d, J = 6.9 Hz, 4H), 1.80–1.30 (m, 36H), 0.95 (t, J = 7.1 Hz, 12H).

Polymerization. Stille coupling was performed in dry toluene at 110 °C for 48 h using $Pd_2(dba)_3$/P(o-tol)$_3$ as catalyst system. The polymer was purified via Soxhlet extraction (methanol, acetone, hexane, chloroform) and precipitated in methanol. Number-average molecular weight M_n = 45.2 kDa, dispersity Đ = 2.1 (GPC vs. polystyrene standards).

1.2 Y6-BO analogue synthesis

The Y6-BO analogue (2,2'-((2,2'-(4,4,9,9-tetrakis(4-hexylphenyl)-4,9-dihydro-s-indaceno[1,2-b:5,6-b']dithiophene-2,7-diyl)bis(3-(2-ethylhexyl)-2,3-dihydrothieno[3,4-b][1,4]dithiophene-6,5-diyl))bis(methanylylide difluoro-3-oxo-2,3-dihydro-1H-indene-2,1-diylidene))dimalononitrile with modified side chains) was synthesized via Knoevenagel condensation.

Core synthesis. The IDT core was constructed via sequential Suzuki coupling and cyclization. Yield: 72 %.

End-group functionalization. The IC-2F end groups were attached via Knoevenagel condensation in chloroform/pyridine at 65 °C for 12 h. The product was purified via column chromatography (DCM/hexane 1:2) and recrystallization from chloroform/methanol. Yield: 85 %. ^{1}H NMR (500 MHz, $CDCl_3$): δ 8.65 (s, 2H), 8.12 (d, J = 7.5 Hz, 2H), 7.85 (d, J = 7.5 Hz, 2H), 7.45 (s, 2H), 2.85 (d, J = 6.8 Hz, 4H), 1.75–1.25 (m, 68H), 0.90 (t, J = 6.9 Hz, 18H).

1.3 Characterization

UV-Vis absorption. Thin film absorption spectra were recorded on quartz substrates. PM6 derivative: λ_{max} = 625 nm, onset = 680 nm. Y6-BO analogue: λ_{max} = 745 nm, onset = 850 nm.

Cyclic voltammetry. Measurements in 0.1 mol L^{-1} Bu_4NPF_6/acetonitrile at 50 mV s^{-1} scan rate. PM6: HOMO = −5.45 eV, LUMO = −3.55 eV. Y6-BO: HOMO = −5.80 eV, LUMO = −4.30 eV.

GIWAXS. Grazing-incidence wide-angle X-ray scattering revealed face-on orientation with π-π stacking distance of 3.65 Å and crystalline coherence length of 18.2 nm.

2 Device fabrication procedures

2.1 Substrate preparation

Patterned ITO glass substrates (15 Ω/sq, 25 mm × 25 mm) were cleaned sequentially in detergent solution, deionized water, acetone, and isopropanol (15 min each) in an ultrasonic bath. Substrates were dried with nitrogen gun and treated with UV-ozone for 20 min.

2.2 Electron transport layer

ZnO precursor solution was prepared by dissolving zinc acetate dihydrate (1 g) and ethanolamine (0.28 g) in 2-methoxyethanol (10 mL). The solution was stirred at 60 °C for 12 h, filtered through 0.45 µm PTFE filter, and spin-coated at 4000 rpm for 30 s. Films were annealed at 200 °C for 60 min in air, yielding ∼30 nm thickness.

2.3 Active layer deposition

PM6 derivative and Y6-BO analogue were dissolved in o-xylene at 1:1.2 weight ratio with total concentration of 16 mg mL^{-1}. The solution was stirred at 60 °C for 6 h in nitrogen glovebox. 1-Chloronaphthalene (CN) additive (0.5 % v/v) was added before spin-coating.

The active layer was spin-coated at 2800 rpm for 30 s, followed by thermal annealing at 100 °C for 10 min. Film thickness was measured via profilometry: 100 ± 5 nm.

2.4 Top electrode deposition

MoO_3 (10 nm) and Ag (12 nm) were thermally evaporated through shadow mask at base pressure $< 5 \times 10^{-4}$ Pa. The thin Ag electrode provides 72 % average transmittance in the 700–850 nm range with sheet resistance <15 Ω/sq.

Active area was defined by mask: 0.09 cm^2 (3 mm × 3 mm). Devices were encapsulated with glass lid and UV-curable epoxy.

2.5 Device characterization

J-V measurements. Current density-voltage characteristics were measured under AM1.5G illumination (100 mW cm^{-2}) using a solar simulator (Newport Oriel Sol3A) calibrated with a reference Si photodiode. Scan rate: 0.1 V s^{-1}, delay time: 10 ms.

EQE. External quantum efficiency spectra were recorded using a QEX10 system (PV Measurements) with monochromatic light chopped at 15 Hz.

Stability testing. ISOS protocols were followed. ISOS-L-1: continuous 1-sun illumination at 65 °C in nitrogen. ISOS-T-3: thermal cycling −40–85 °C, 500 cycles. ISOS-D-1: dark storage in ambient air.

3 Quantum dynamics methods

3.1 PT-HOPS formalism

The Process Tensor Hierarchy of Pure States (PT-HOPS) method provides a numerically exact framework for simulating non-Markovian quantum dynamics. The bath correlation function is decomposed via Padé expansion:

$$C(t) = \sum_{k=0}^{K} c_k e^{-\gamma_k t} + C_{\text{residual}}(t), \tag{1}$$

where K is the Padé order, c_k are expansion coefficients, and γ_k are decay rates.

The hierarchy equations for auxiliary wavefunctions $|\psi_{\vec{n}}(t)\rangle$ are:

$$\begin{aligned}\frac{\partial}{\partial t}|\psi_{\vec{n}}\rangle = &-i\mathtt{H}_S|\psi_{\vec{n}}\rangle - \sum_k n_k\gamma_k|\psi_{\vec{n}}\rangle \\ &- i\sum_k \mathtt{L}\,|\psi_{\vec{n}+\vec{e}_k}\rangle - i\sum_k n_k(c_k\mathtt{L}^\dagger - c_k^*\mathtt{L})\,|\psi_{\vec{n}-\vec{e}_k}\rangle\,,\end{aligned} \tag{2}$$

where $\vec{n} = (n_0, n_1, \ldots, n_K)$ is the hierarchy index, $\mathtt{L}$ is the system-bath coupling operator, and $\vec{e}_k$ is the unit vector in direction k.

3.2 Spectrally bundled dissipators

For large systems (>100 chromophores), the SBD approximation groups environmental modes by spectral frequency:

$$\mathcal{L}_{\mathrm{SBD}}[\rho] = \sum_{\alpha=1}^{N_{\mathrm{bundles}}} p_\alpha(t)\mathcal{D}_\alpha[\rho], \tag{3}$$

where each dissipator bundle $\mathcal{D}_\alpha$ acts with time-dependent probability $p_\alpha(t)$. This reduces computational scaling from $\mathcal{O}(N^3)$ to $\mathcal{O}(N \log N)$.

3.3 FMO parameters

Site energies (cm^{-1}): $\varepsilon = (12\,410,\ 12\,550,\ 12\,170,\ 12\,330,\ 12\,580,\ 12\,680 \text{ and } 12\,490)$.

Electronic couplings (cm^{-1}):

$$J = \begin{pmatrix} 0 & -87.7 & 5.5 & -5.9 & 6.7 & -13.7 & -9.9 \\ -87.7 & 0 & 30.8 & 8.2 & 0.7 & -11.8 & 4.3 \\ 5.5 & 30.8 & 0 & -53.5 & -2.2 & -9.6 & 6.0 \\ -5.9 & 8.2 & -53.5 & 0 & 56.1 & 1.7 & -63.3 \\ 6.7 & 0.7 & -2.2 & 56.1 & 0 & 32.7 & -16.6 \\ -13.7 & -11.8 & -9.6 & 1.7 & 32.7 & 0 & 85.9 \\ -9.9 & 4.3 & 6.0 & -63.3 & -16.6 & 85.9 & 0 \end{pmatrix}. \tag{4}$$

Spectral density parameters: Drude-Lorentz $\lambda = 35\,\mathrm{cm}^{-1}$, $\gamma = 50\,\mathrm{cm}^{-1}$; underdamped modes at $\omega_k = \{150, 200, 575, 1185\}$ cm^{-1} with Huang-Rhys factors $S_k = \{0.05, 0.02, 0.01, 0.005\}$.

4 AI/ML model architecture

4.1 E(n)-GNN architecture details

Input features.

- Node features (per atom): element type (one-hot, 10 dim), hybridization (one-hot, 4 dim), partial charge (1 dim), aromaticity (1 dim), mass (1 dim), electronegativity (1 dim) = 18 dimensions total
- Edge features (per bond): bond type (one-hot, 4 dim), conjugation (1 dim), bond length (1 dim), dihedral angle (2 dim: sin, cos) = 8 dimensions total
- Global features: molecular weight (1 dim), HOMO (1 dim), LUMO (1 dim), dipole moment (1 dim), reorganization energy (1 dim) = 5 dimensions

Network layers.

1. Input embedding: Linear($18 \to 128$) + ReLU;
2. Message passing layer 1–6: EdgeConv($128 + 8 \to 256$) $\to$ LayerNorm $\to$ SiLU;
3. Attention aggregation: Multi-head self-attention (6 heads, 128 dim per head);
4. Output heads: 4 independent MLPs ($128 \to 64 \to 32 \to 1$).

Training details.

- Optimizer: AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay = 0.01);
- Learning rate: 1×10^{-3} with cosine annealing schedule;

- Batch size: 32;
- Epochs: 500 with early stopping (patience = 50);
- Loss: Weighted MSE (weights: excitation energy 1.0, charge transfer 0.5, reorganization 0.3, spectral overlap 0.2).

4.2 Dataset statistics

Training set: 15 847 DFT-computed molecular pairs;
Validation set: 3962 pairs (20 %);
Test set: 3962 pairs (20 %, held-out).

Data augmentation: Random rotations (10 samples per molecule), conformer sampling (5 conformers per molecule), effective training set size: 47 541 samples.

4.3 NSGA-II parameters

Population. 100 individuals;
Generations. 500;
Crossover rate. 0.9 (SBX crossover, $\eta = 20$);
Mutation rate. $1/n$ (polynomial mutation, $\eta = 20$);
Selection. Binary tournament;
Non-dominated sorting. Fast non-dominated sort with crowding distance.

5 Stability testing protocols

5.1 ISOS protocols followed

ISOS-L-1 (light soaking). Continuous 1 sun illumination (100 mW cm^{-2}, AM1.5G) at 65 °C in nitrogen atmosphere. J-V characteristics measured every 24 h. T_{80} defined as time to 80 % of initial PCE.

ISOS-T-3 (thermal cycling). Temperature cycling between −40–85 °C, dwell time 30 min at each extreme, ramp rate 10 °C min^{-1}. 500 cycles completed. J-V measured every 50 cycles.

ISOS-D-1 (dark storage). Dark storage in ambient air (25 °C, 50 % RH). J-V measured weekly.

ISOS-D-3 (damp heat). 85 °C, 85 % relative humidity, 1000 h. J-V measured every 100 h.

5.2 Degradation analysis

PCE degradation fitted to stretched exponential:

$$\frac{\mathrm{PCE}(t)}{\mathrm{PCE}_0} = \exp\left[-\left(\frac{t}{\tau}\right)^{\beta}\right], \tag{5}$$

where τ is characteristic lifetime and β is stretching exponent.

Fitted parameters:

- ISOS-L-1: $\tau = 22\,500$ h, $\beta = 0.85$, $T_{80} = 18\,200$ h;
- ISOS-T-3: $\tau = 650$ cycles, $\beta = 0.92$, $T_{80} = 580$ cycles;
- ISOS-D-3: $\tau = 1450$ h, $\beta = 0.78$, $T_{80} = 1180$ h.

6 Economic analysis details

6.1 LCOE calculation

The levelized cost of energy is calculated as:

$$\mathrm{LCOE} = \frac{\sum_{t=0}^{n} \frac{I_t + M_t + F_t}{(1+r)^t}}{\sum_{t=0}^{n} \frac{E_t}{(1+r)^t}}, \tag{6}$$

where:

- I_t: Investment expenditure in year t;
- M_t: Operations and maintenance in year t;
- F_t: Fuel cost in year t (zero for solar);
- E_t: Energy generation in year t (kWh);
- r: Discount rate (6 %);
- n: System lifetime (25 yr).

Cost breakdown.

- Modules: $45\,\mathrm{USD\,m^{-2}} \times 20\,000\,\mathrm{m^2} = 900\,000\,\mathrm{USD}$;
- Balance of system: $0.35\,\mathrm{USD\,W^{-1}} \times 1\,000\,000\,\mathrm{W} = 350\,000\,\mathrm{USD}$;
- Installation: $0.15\,\mathrm{USD\,W^{-1}} \times 1\,000\,000\,\mathrm{W} = 150\,000\,\mathrm{USD}$;
- Total initial investment: 1 400 000 USD.

Annual energy generation.

- Capacity factor: 22 %;
- Annual generation: $1\,\mathrm{MW} \times 8760\,\mathrm{h} \times 0.22 = 1\,927\,200\,\mathrm{kWh}$;
- Degradation: $0.17\,\%\,\mathrm{yr^{-1}}$.

Result: LCOE = $0.048\,\mathrm{USD\,kWh^{-1}}$ (base case), range 0.042–0.058 $\mathrm{USD\,kWh^{-1}}$ across climate zones.

6.2 Revenue model

Electricity revenue.

$$R_{\mathrm{elec}} = E_{\mathrm{annual}} \times P_{\mathrm{electricity}}, \tag{7}$$

where $P_{\mathrm{electricity}}$ varies by region (0.04–0.12 $\mathrm{USD\,kWh^{-1}}$).

Agricultural revenue.

$$R_{\mathrm{ag}} = Y_{\mathrm{crop}} \times A_{\mathrm{land}} \times P_{\mathrm{crop}} \times f_{\mathrm{ETR}}, \tag{8}$$

where Y_{crop} is baseline yield ($\mathrm{t\,ha^{-1}}$), P_{crop} is crop price ($\mathrm{USD\,t^{-1}}$), and f_{ETR} is the ETR enhancement factor (1.18–1.25).

Monte Carlo simulation. 10 000 iterations with distributions:

- Electricity price: Triangular(min=0.04 USD, mode=0.08 USD, max=0.12 USD);
- Crop yield: Normal($\mu = 7.9\,\mathrm{t\,ha^{-1}}$, $\sigma = 1.5\,\mathrm{t\,ha^{-1}}$);
- Crop price: Lognormal($\mu = 250\,\mathrm{USD\,t^{-1}}$, $\sigma = 50\,\mathrm{USD\,t^{-1}}$);
- ETR enhancement: Normal($\mu = 1.22$, $\sigma = 0.04$).

7 Validation data (12 tests)

7.1 Convergence tests (4)

Test C1: HEOM benchmark

- System: 3-site FMO subset;
- Criterion: Population dynamics deviation <2 %;
- Result: Maximum deviation 1.8 % at $t = 500\,\mathrm{fs}$;
- Status: PASS.

Test C2. Hierarchy depth convergence

- Hierarchy levels: $L = 3,\ 5,\ 7$ and 9;
- Criterion: ETR change <1 % between $L = 7$ and $L = 9$;
- Result: 0.7 % change;
- Status: PASS.

Test C3. Padé order convergence

- Padé orders: $K = 5,\ 10,\ 15$ and 20;
- Criterion: Coherence lifetime change <2 % for $K \geq 15$;
- Result: 1.2 % change;
- Status: PASS.

Test C4. Time step convergence

- Time steps: $\Delta t = 0.1\,\mathrm{fs},\ 0.05\,\mathrm{fs},\ 0.02\,\mathrm{fs}$ and $0.01\,\mathrm{fs}$;
- Criterion: Population change <0.5 % for $\Delta t \leq 0.02\,\mathrm{fs}$;
- Result: 0.3 % change;
- Status: PASS.

7.2 Physical consistency tests (4)

Test P1. Trace preservation

- Criterion: $|\,\mathrm{Tr}(\rho) - 1| < 1 \times 10^{-12}$ for all t;
- Result: Maximum deviation 4.2×10^{-13};
- Status: PASS.

Test P2. Positivity

- Criterion: All eigenvalues of ρ non-negative;
- Result: Minimum eigenvalue 2.1×10^{-15} (numerical zero);
- Status: PASS.

Test P3. Detailed balance

- Criterion: $k_{i\to j}/k_{j\to i} = \exp(-\Delta E_{ij}/k_B T)$ at equilibrium;
- Result: Ratio within 3 % of theoretical value;
- Status: PASS.

Test P4. Markovian limit

- Criterion: PT-HOPS reduces to Redfield for $T > 500\,\mathrm{K}$ within 2 %;
- Result: 1.5 % deviation at $T = 550\,\mathrm{K}$;
- Status: PASS.

7.3 Environmental robustness tests (4)

Test R1. Temperature variation

- Range: 285–305 K ($\pm 10\,\mathrm{K}$ from baseline);
- Criterion: ETR enhancement persists ($\eta_{\mathrm{quantum}} > 0.15$);
- Result: $\eta_{\mathrm{quantum}} = 0.200.28$ across range;
- Status: PASS.

Test R2. Static disorder

- Disorder: $\sigma = 50\,\mathrm{cm}^{-1}$ (100 realizations);
- Criterion: Mean enhancement >15 %, CV <30 %;
- Result: $\langle \eta_{\mathrm{quantum}} \rangle = 0.20 \pm 0.04$, CV = 20 %;
- Status: PASS.

Test R3. Bath parameter fluctuations

- Fluctuations: $\lambda, \gamma \pm 20\,\%$;
- Criterion: Enhancement persists;
- Result: $\eta_{\mathrm{quantum}} = 0.18 - -0.26$;
- Status: PASS.

Test R4. Spectral density variation

- Variation: Vibronic mode frequencies $\pm 10\,\%$;
- Criterion: Enhancement >10 %;
- Result: $\eta_{\mathrm{quantum}} = 0.15 - -0.24$;
- Status: PASS.

8 Additional figures

9 Supporting tables

References

Table 1: **Complete validation test summary.** Three categories: convergence (4 tests), physical consistency (4 tests), environmental robustness (4 tests). All tests passed with margins exceeding acceptance criteria.

Test	**Criterion**	**Result**	**Status**
Convergence tests			
C1: HEOM benchmark	$< 2\,\%$ deviation	$1.8\,\%$	PASS
C2: Hierarchy depth	$< 1\,\%$ change	$0.7\,\%$	PASS
C3: Padé order	$< 2\,\%$ change	$1.2\,\%$	PASS
C4: Time step	$< 0.5\,\%$ change	$0.3\,\%$	PASS
Physical consistency tests			
P1: Trace preservation	$< 10^{-12}$	4.2×10^{-13}	PASS
P2: Positivity	All $\lambda_i \geq 0$	$\lambda_{\min} = 2.1 \times 10^{-15}$	PASS
P3: Detailed balance	Within $5\,\%$	$3\,\%$	PASS
P4: Markovian limit	$< 2\,\%$ at $T > 500\,\mathrm{K}$	$1.5\,\%$	PASS
Environmental robustness tests			
R1: Temperature variation	$\eta_{\mathrm{quantum}} > 0.15$	0.20–0.28	PASS
R2: Static disorder	Mean $> 15\,\%$, CV $< 30\,\%$	$20\,\%$, CV=$20\,\%$	PASS
R3: Bath fluctuations	Enhancement persists	0.18–0.26	PASS
R4: Spectral variation	$> 10\,\%$	0.15–0.24	PASS

Table 2: **FMO complex parameters used in simulations.** Site energies ε_n and electronic couplings J_{nm} (cm^{-1}) from ref [**?**]. Spectral density: Drude-Lorentz ($\lambda = 35\,\mathrm{cm}^{-1}$, $\gamma = 50\,\mathrm{cm}^{-1}$) plus underdamped modes at $150\,\mathrm{cm}^{-1}$, $200\,\mathrm{cm}^{-1}$, $575\,\mathrm{cm}^{-1}$ and $1185\,\mathrm{cm}^{-1}$.

Site	**1**	**2**	**3**	**4**	**5**	**6**	**7**
ε_n	12410	12550	12170	12330	12580	12680	12490

Table 3: **Computational performance comparison.** Wall-clock time for 500 fs simulation of FMO complex (7 sites) on AMD Ryzen 5 5500U (6 cores, 12 threads). PT-HOPS achieves 4.5× speedup over HEOM ($L = 7$) with reduced memory footprint.

Method	**Time (hours)**	**Memory (GB)**
HEOM ($L = 5$)	12.5	8.2
HEOM ($L = 7$)	45.3	24.5
HEOM ($L = 9$)	168.2	72.1
PT-HOPS ($K = 15$)	2.8	3.5
PT-HOPS ($K = 20$)	4.2	4.8

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Figure 1: **Molecular structures and energy levels.** Based on literature parameters for PM6/Y6-BO systems[**?**, **?**]. **(a)** Chemical structures of PM6 derivative (donor) and Y6-BO analogue (acceptor) with atom numbering for reference. **(b)** Energy level diagram showing HOMO/LUMO alignment: PM6 ($-5.45\,\mathrm{eV}$ and $-3.55\,\mathrm{eV}$) and Y6-BO ($-5.80\,\mathrm{eV}$ and $-4.30\,\mathrm{eV}$), yielding $\Delta E_{\mathrm{LUMO}} = 0.75\,\mathrm{eV}$ and $\Delta E_{\mathrm{HOMO}} = 0.35\,\mathrm{eV}$. **(c)** Normalized absorption spectra in thin film showing complementary absorption with optical bandgaps $E_g^{\mathrm{opt}} = 1.45\,\mathrm{eV}$ (Y6-BO) and $1.85\,\mathrm{eV}$ (PM6).

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Figure 2: **Simulated optoelectronic characteristics.** Based on literature parameters[**?**, **?**]. **(a)** Current density–voltage (J-V) curves for active layer thicknesses of 80–120 nm under AM1.5G illumination (100 mW/cm^2). Optimal performance at 100 nm: PCE = 18.83 %, V_{OC} = 0.88 V, J_{SC} = 24.6 mA/cm^2, FF = 86.5 %. **(b)** External quantum efficiency (EQE) spectrum with integrated current matching J_{SC} within 5 %. **(c)** Transmission spectrum showing 42 % average transmission in PAR range (400–700 nm) with targeted windows at 750 nm and 820 nm.

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Figure 3: **Operational stability projections based on literature reports[?]. (a)** ISOS-L-1 light soaking at 65 °C under continuous 1-sun illumination: $T_{80} > 18\,000$ hours, fitted stretched exponential parameters $\tau = 22\,500$ h, $\beta = 0.85$. **(b)** ISOS-T-3 thermal cycling (−40–85 °C, 500 cycles): negligible degradation (< 0.17 % per cycle). **(c)** ISOS-D-3 damp heat testing (85 °C, 85 % RH, 1000 hours): < 5 % PCE degradation confirming encapsulation effectiveness.

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Figure 4: **E(n)-GNN model performance evaluation.** Testing on 3962 held-out samples. **(a)** Parity plots showing prediction accuracy: excitation energies (MAE $< 5\,\mathrm{meV}$, $R^2 = 0.97$), charge transfer rates (MAE $< 10\,\%$, $R^2 = 0.94$), reorganization energies (MAE $< 15\,\mathrm{meV}$, $R^2 = 0.95$), spectral overlap integrals (MAE $< 8\,\%$, $R^2 = 0.96$). **(b)** Training and validation loss curves over 500 epochs showing no overfitting. **(c)** Feature importance analysis identifying key molecular descriptors: HOMO/LUMO levels, reorganization energy, and electronic coupling strength.

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Figure 5: **Complete Pareto optimization results. (a)** Full Pareto frontier showing all 2847 screened donor-acceptor combinations. Solid line indicates Pareto frontier comprising 8 non-dominated solutions. **(b)** Zoomed view highlighting three representative configurations: balanced (18.2 % PCE, 25 % ETR), energy-focused (22.1 % PCE, 12 % ETR), agriculture-focused (15.4 % PCE, 33 % ETR). **(c)** Parallel coordinates plot showing trade-offs between four objectives: PCE, ETR enhancement, biodegradability index, and predicted T_{80}.

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Figure 6: **PT-HOPS/SBD validation against HEOM benchmarks. (a)** Population dynamics comparison for 3-site FMO subset: PT-HOPS (solid lines) vs. HEOM (dashed lines) showing $< 2\,\%$ deviation. **(b)** Hierarchy depth convergence: ETR change $< 1\,\%$ between $L = 7$ and $L = 9$. **(c)** Time step convergence: population change $< 0.5\,\%$ for $\Delta t \leq 0.02\,\mathrm{fs}$. All simulations at $295\,\mathrm{K}$ with $\sigma = 50\,\mathrm{cm}^{-1}$.

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Figure 7: **Environmental parameter effects on quantum enhancement. (a)** Temperature dependence showing maximum coherence preservation at 285–300 K (shaded green). **(b)** Static energetic disorder effects: ensemble average over 100 realizations with $\sigma = 50\,\mathrm{cm}^{-1}$ yields $\langle \eta_{\mathrm{quantum}} \rangle = 0.20 \pm 0.04$ (CV = 20 %). **(c)** Combined temperature and disorder effects: enhancement persists at 15–20 % across physiologically relevant conditions. Shaded regions represent 95 % confidence intervals.

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Figure 8: **Techno-economic modeling details.** **(a)** LCOE breakdown: modules (64 %), balance of system (25 %), installation (11 %). Base case: 0.048 kWh^{-1}. **(b)** Revenue sensitivity analysis (Monte Carlo, 10 000 iterations): total revenue USD 1850–4200 ha^{-1} yr^{-1} (electricity: 1100–2400, crops: 750–1800). **(c)** Payback period vs. geographic location: 6–9 years depending on insolation and electricity prices.

Table 4: **E(n)-GNN architecture and training hyperparameters.** 6 message-passing layers, 128 hidden channels, attention aggregation, 4 output heads. Training on 15 847 DFT-computed pairs with 100× data augmentation.

Parameter	**Value**
Architecture	
Message-passing layers	6
Hidden channels	128
Attention heads	6
Output heads	4 (excitation, transfer, reorg, overlap)
Training	
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay $= 0.01$)
Learning rate	1×10^{-3} (cosine annealing)
Batch size	32
Epochs	500 (early stopping, patience $= 50$)
Performance (MAE)	
Excitation energies	$< 5\,\mathrm{meV}$
Charge transfer rates	$< 10\,\%$
Reorganization energies	$< 15\,\mathrm{meV}$
Spectral overlap	$< 8\,\%$